gpeyt1-44 · 2:1 stress + clay settlement

GEO PE · Long Calculation Problems I — Topic 9: Shallow Foundations

Problem

A 10 ft × 10 ft footing with net applied pressure Δp = 2,600 psf bears 3 ft below grade on sand (γ = 120 pcf above and below the water table, which is at the footing base). A 10 ft normally consolidated clay layer (γsat = 112 pcf, e0 = 0.92, Cc = 0.3) begins 10 ft below the footing base. Use the 2:1 method at mid-clay and a single layer.

Most nearly, the consolidation settlement of the clay is:

A. 9.5 in.
B. 3.6 in.
C. 2.5 in.
D. 1.5 in.

Given equations / data

> $\Delta\sigma_z = \Delta p\, B\, L/[(B + z)(L + z)]$ (2:1, z below footing base); $S = C_c H/(1 + e_0) \log(\sigma'_f/\sigma'_0)$; $\gamma_w = 62.4$ pcf

Worked solution

1. z (base → mid-clay) = 10 + 5 = 15 ft
2. $\Delta\sigma = 2{,}600 \times 10^2/(10+15)^2 = 416$ psf
3. $\sigma'_0 = 3(120) + 10(120-62.4) + 5(112-62.4) = 1{,}184$ psf
4. $S = 0.3 \times 10/1.92 \times \log(1{,}600/1{,}184) = 0.2043$ ft = 2.5 in.

Final answer

2.5 in.

Why the other values are wrong

- **9.5 in.** — applied full Δp at mid-clay (no spreading)
- **3.6 in.** — evaluated Δσ at the top of clay
- **1.5 in.** — used total stress for σ′0

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.5.1 Stress Distribution (2:1 method), p. 94; §3.2.1 NC consolidation, p. 83

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=NwGztxby-vQ, https://www.youtube.com/watch?v=SLtD_6vcRAs (no creator values copied).

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